

Blink Qt — screen sharing as a video device

Status as of 2026-10-09: findings and proposed design, nothing implemented yet.

Why

Blink Qt shares the desktop over VNC: `x11vnc` captures the screen and the RFB stream travels over MSRP. `x11vnc` only works on an X11 session.

- Ubuntu 26.04 (resolute) ships GNOME with no Xorg session (dropped in 25.10), so sharing *out* fails there. Viewing a remote screen with the built-in RFB viewer still works.
- On noble, bookworm and trixie the default GNOME session is already Wayland; users can still choose an X11 session, but that option is going away.

What Blink for macOS does

Screen sharing is a video capture device, implemented in python3-sipsimple as `deps/patches/2.17/32_avf_screen_capture.patch`:

- `avf_dev.m` lists every active display after the cameras: “My screen” (main display), “My screen 2”, ...
- Picking one as the camera sends the screen instead of the camera. Nothing changes above pjmedia: the call is an ordinary video call, so it interoperates with any client (Sylk Mobile, WebRTC, other SIP phones).
- Capture uses ScreenCaptureKit (macOS 12.3+), weak-linked; older systems list no screen devices.
- Needs the Screen Recording permission. Without it the first open triggers the system prompt and fails with `PJMEDIA_EVID_NOTREADY`, and the application falls back to a camera.
- ScreenCaptureKit only delivers a frame when the screen changes, while the packetizer advances the RTP timestamp by a fixed amount per frame. The callback keeps only the latest buffer and a timer at the configured frame rate hands it to pjmedia. This keeps timing right and lets keyframe requests be served on a static screen.
- Output is BGRA at the requested size, letterboxed.

Proposed Linux design

The same shape: a new pjmedia capture device next to `v4l2_dev.c`, in python3-sipsimple, exposed as “My screen” in the camera list.

Capture backend

1. **xdg-desktop-portal ScreenCast + PipeWire** (primary). The only method that works on Wayland; GNOME and KDE also offer it on X11 sessions.
 - D-Bus handshake with `org.freedesktop.portal.ScreenCast`: `CreateSession` → `SelectSources` → `Start` → `OpenPipeWireRemote`, giving a PipeWire fd and node id.
 - Consume the node with libpipewire. Ask for shared-memory buffers (`MemFd/MemPtr`, `BGRx/RGBx`), not `DMA-BUF`, to avoid GPU import.
2. **XShm on X11** (optional fallback) for sessions without a portal backend, e.g. bare window managers. Can list monitors through XRandR the way macOS lists displays.

Carried over from the macOS patch

- Keep only the latest PipeWire buffer; a timer at the configured frame rate hands it to pjmedia (same reason as ScreenCaptureKit: PipeWire delivers on damage only).
- Scale and letterbox to the requested size with `libswscale` (already linked through `ffmpeg`).
- First open returns `PJMEDIA_EVID_NOTREADY` while the user is being asked, and Blink falls back to the camera.

Differences from macOS

	macOS	Linux (portal)
Permission	Screen Recording, granted once in System Settings	Portal picker; restore token (ScreenCast v4, <code>persist_mode=2</code>) skips it on later calls
Device list	One device per display	One “My screen”; the picker chooses the monitor (or the XShm fallback lists monitors)
What can be shared	Whole display	Whole monitor or a single window
First use	Fails with NOTREADY, system prompt	Waits on the picker; return NOTREADY and complete in the background

The restore token has to be stored, e.g. as a python3-sipsimple setting, so later calls go straight through.

Changes needed

python3-sipsimple:

- New `screen_dev.c` in `pjmedia-videodev` (roughly the size of patch 32), as a new numbered patch in `deps/patches/2.17`
- Makefile / `aconfigure` gate and a `PJMEDIA_VIDEO_DEV_HAS_...` define in `setup_pjsip.py` (Linux only)
- Build-Depends: `libpipewire-0.3-dev`, `libglib2.0-dev` (GDBus for the portal); for the XShm fallback also `libx11-dev`, `libxext-dev`, `libxrandr-dev`. All available on bookworm, trixie, noble and resolute.

Blink Qt:

- Little or nothing: the device appears in the camera list on its own
- Fall back to the camera on `PJMEDIA_EVID_NOTREADY`
- Store the portal restore token
- Keep `x11vnc` and VNC desktop sharing for remote control on X11; `screen-as-video` is the way to share on Wayland

Runtime: PipeWire and a portal backend (`xdg-desktop-portal-gnome`, `-kde` or `-wlr`), which the desktop already provides.

Open decision

1. Portal + PipeWire only (recommended)
2. Portal + PipeWire with the XShm fallback